

Algorithms and Data Structures

Algorithms are step-by-step procedures for solving problems, and data structures organize information for efficient use.

Key facts:

A data structure stores and organizes data to support operations such as search, insertion, deletion, and traversal.

Arrays provide indexed access, while linked lists make some insertions and deletions easier when node references are known.

Stacks follow last-in, first-out order; queues follow first-in, first-out order.

Hash tables use a hash function to map keys to storage locations and often support average-case constant-time lookup.

Trees represent hierarchical relationships, and binary search trees keep keys ordered to support efficient search.

Graphs model relationships among nodes and edges, such as networks, routes, dependencies, and social connections.

Big O notation describes how resource use grows with input size, abstracting away machine-specific constants.

Answerable questions:

Q: What is Big O notation? A: Big O describes how an algorithm's time or space use grows as input size increases.

Q: What is a hash table used for? A: It maps keys to values and is commonly used for fast lookup, insertion, and deletion.